GEARBOx
LLM Run 3 Review

August 30, 2024

Run 3

- Hierarchical approach

  Map GEARBOx variables with two separate prompts capturing inclusion and exclusion criteria

- Performance

	NCT04898894	NCT04915612	NCT05188170
Incorrect values	IC: 4 / 112 EC: 0 / 112	IC: 1 / 112 EC: 5 / 112	IC: 2 / 112 EC: 8 / 112
Missed array values	IC: 2 / 2 EC: N/A	IC: 2 / 2 EC: N/A	IC: N/A EC: 1 / 1 (CNS)

Findings - Null values

- Improvement: Consistent null values
  - “None”: numeric type variable without Enum values
  - “Not applicable”: string type variable with Enum values

- Mixed output
  - “None” for string type variable with Enum values; and
  - “Not applicable” for numeric type variable without Enum values; or vice versa
  - Cases included: ‘None’:
    NCT04898894:18-cns_status, 110-expression_alk_type, 111-bclc_stage
    NCT04915612: 18-cns_status, 108-cumul_131I_mibg, 110-expression_alk_type, 111-bclc_stage
    NCT05188170: 52-type_transplant, 64-cumul_anthracycline, 108-cumul_131I_mibg, 110-expression_alk_type, 111-bclc_stage
  - Cases included: ‘Not Applicable’:
    NCT04898894:52-type_transplant

Findings

- EC prompt captures “Yes” for exclusion. Need to review wordings on GEARBOx question texts for clarification.

- Aggregation of variables:
  70-exp_venetoclax: Does the eligibility criteria mention venetoclax?
  LLM output: IC: “Yes”; EC: “Yes”
  Expected values: Yes (‘Yes’ for inclusion); No (‘Yes’ for exclusion)

- Decision: No aggregation necessary.

Findings - Extracting values from subtle context

- Difficulty in extracting values from subtle context
  - EC does not mention variable explicitly
    - e.g.) sex
      NCT04898894: Male or female of reproductive potential must agree to use effective contraception for the duration of study participation.

  - Variables appear in various form in eligibility criteria
    - Knowledgebase representation: Examples exist, No Enum exists.
      e.g.) 18-cns_status: What CNS status is mentioned in the eligibility criteria?
      Examples: "CNS 1", "CNS 2", "CNS 3"
      111-bclc_stage: What BCLC stage is mentioned in the eligibility criteria?
      Examples: "Stage 0", "Stage A", "Stage B", "Stage C", "Stage D"

Findings - Variable type change considerations

- Potential **array** type variables
  - Issue: LLMs do not always produce multi-value (or array) output for string type variables.

Variable	Criteria text	LLM Output	Expected Output
7-diagnosis	Participants must have a diagnosis of AML or ALAL	"Acute myeloid leukemia (AML)"	'Acute myeloid leukemia (AML)', 'Ambiguous lineage acute leukemia (ALAL)'
14-bm_recent_blast_method: What are the required methods for bone marrow blast percentage assessment mentioned in the eligibility criteria?	Patients must have >= 5% blasts in the bone marrow as assessed by morphology or flow cytometry	Morphology	Morphology or flow cytometry

* Example is from NCT04898894.

Findings - Variable type change considerations

- Numeric vs. string types
  - Units and measurements:
    - String type can capture greater than, less than, and equal to signs.
      E.g.) 64-cumul_anthracycline:
      "450" OR "<= 450 mg/m^2"

    - More examples: 10-current_refractory_num_cycles, 12-count_relapse, 13-bm_recent_blast_pct_meas, 16-periph_blood_blast_pct_meas

Discussion

- Conditional variables or washout-related variables
  - E.g.) variables start with “exp_”
    51-exp_hct: Does the eligibility criteria mention any prior exposure to Hematopoietic cell transplantation (HCT)?
    52-type_transplant: What type(s) of transplant are mentioned in the eligibility criteria?
  - Capture supporting text or related variable(s) on output

- Status of multi-run approach